Table 1: Performance of stock trading with different LLMs as backbone model across seven stocks.

Model	MSFT				JNJ				UVV			
	CR↑	SR↑	AV↓	MDD↓	CR↑	SR↑	AV↓	MDD↓	CR↑	SR↑	AV↓	MDD↓
Buy & Hold	15.340	1.039	24.980	9.428	13.895	1.343	17.500	9.847	36.583	2.112	29.299	15.406
Financial Domain Models												
Palmyra-Fin-70B	14.697	0.897	27.518	9.428	5.748	0.450	19.317	9.367	37.875	2.039	31.200	15.967
Proprietary Models												
GPT-o1-preview	17.184	0.962	30.000	9.428	13.561	1.086	20.864	9.847	41.508	2.147	32.479	9.633
GPT-4	16.654	0.932	30.022	9.428	13.712	1.103	20.894	9.860	31.791	1.640	32.567	10.434
GPT-4o	12.461	0.924	22.653	6.647	9.099	0.875	17.471	7.169	8.043	0.496	27.241	14.889
Open-Source Models												
Qwen2.5-72B-Instruct	7.421	0.588	21.238	6.973	14.353	1.140	20.995	9.812	37.178	1.822	34.223	13.365
Llama-3.1-70B-Instruct	17.396	1.335	21.892	7.045	13.868	1.121	20.779	9.825	35.981	1.728	34.986	15.406
DeepSeek-67B-Chat	13.941	0.834	28.081	7.850	14.426	1.185	20.450	9.825	29.940	1.481	33.964	15.407
Yi-1.5-34B-Chat	22.093	1.253	29.613	9.428	14.004	1.180	19.938	9.847	20.889	1.020	34.417	14.936
Qwen2.5-32B-Instruct	-0.557	-0.041	22.893	8.946	2.905	0.292	16.725	7.169	-1.623	-0.097	27.973	17.986
DeepSeek-V2-Lite (15.7B)	11.904	0.694	28.796	16.094	-7.482	-0.670	18.773	17.806	33.560	1.703	33.099	12.984
Yi-1.5-9B-Chat	19.333	1.094	29.690	9.428	18.606	1.611	19.409	10.986	49.415	2.410	34.446	11.430
Llama-3.1-8B-Instruct	22.703	1.322	28.855	7.385	13.988	1.486	20.460	9.969	41.108	1.981	34.866	16.429
Qwen-2.5-Instruct-7B	-10.305	-0.724	23.937	23.371	21.852	0.980	37.425	9.573	11.752	0.853	22.988	15.451
FLAG-TRADER												
SmolLM2-135M-Instruct	20.106	1.373	24.932	9.428	33.724	3.344	17.174	9.320	46.799	1.463	67.758	35.039

[1] The Buy & Hold strategy is a passive investment approach commonly used as a baseline strategy, where an investor purchases stocks and holds onto them for an extended period regardless of market fluctuations.
[2] An upward arrow (↑) next to a metric indicates that higher values signify better performance, while a downward arrow (↓) indicates that lower values are preferable.
[3] The numbers highlighted in red indicate the best-performing outcomes for the corresponding metrics.

Small-scale models (under 10B parameters) run on two RTX A6000 GPUs (48GB each), mid-scale models (10B–65B parameters) require four RTX A6000 GPUs, and large-scale models (over 65B parameters) use eight A100 GPUs (80GB each). These setups provide sufficient resources for both inference and training, enabling a fair comparison of trading performance across different assets. FLAG-TRADER is trained by using PPO algorithm, which is detailed in Algorithm 2 in Appendix A.

5.2 Evaluation Metrics

We use four widely recognized financial metrics (Hull, 2007) to evaluate and compare the investment performance of various LLM backbones across different tasks: Cumulative Return (CR), Sharpe Ratio (SR), Annualized Volatility (AV), and Maximum Drawdown (MDD). As CR and SR focus more on long-term gains and risk-adjusted returns, they are typically considered more important than AV and MDD for assessing asset trading performance. Accordingly, we treat CR and SR as our primary metrics for the final evaluation.

Cumulative Return (CR) % measures the total value change of an investment over time by summing logarithmic return calculated from daily PnL:

$$\textbf{CR} = \sum_{t=1}^{T} \log\left(1 + \frac{pnl_t}{C_{t-1} + H_{t-1}P_{t-1}}\right), \quad (13)$$

where pnl_t is the PnL at time t, and $C_{t-1} + H_{t-1}P_{t-1}$ is the account balance at time $t-1$. Notice that higher values indicate better strategy effectiveness.

Sharpe Ratio (SR) assesses risk-adjusted returns by dividing the average excess return (R_p) over the risk-free rate (r_f) by its volatility (σ_p):

$$\textbf{SR} = \frac{R_p - r_f}{\sigma_p}. \quad (14)$$

Notice that higher ratios signify better performance.

Annualized Volatility (AV) % and Daily Volatility (DV) % quantify return fluctuations; AV is derived by scaling DV (*standard deviation of daily logarithmic returns*) by the square root of the annual trading days (252):

$$\textbf{AV} = \textbf{DV} \times \sqrt{252}. \quad (15)$$

This metric highlights potential return deviations across the year.

Max Drawdown (MDD) % calculates the largest drop from peak to trough of the value of balance account:

$$\textbf{MDD} = \max\left(\frac{V_{\text{peak}} - V_{\text{trough}}}{V_{\text{peak}}}\right). \quad (16)$$

Notice that lower values indicate lesser risk and higher strategy robustness.

Table 2: Performance of stock trading with different LLMs as backbone model across seven stocks.

Model	HON				TSLA				BTC			
	CR↑	SR↑	AV↓	MDD↓	CR↑	SR↑	AV↓	MDD↓	CR↑	SR↑	AV↓	MDD↓
Buy & Hold	33.256	2.347	23.967	9.195	39.244	0.869	75.854	37.975	21.821	0.683	37.426	20.796
Financial Domain Models												
Palmyra-Fin-70B	20.016	1.464	22.974	6.824	-6.661	-0.222	50.379	25.820	-20.812	-1.212	20.036	27.782
Proprietary Models												
GPT-o1-preview	13.162	0.776	28.511	11.558	34.499	0.796	72.822	35.490	34.060	1.114	35.846	17.075
GPT-4	34.342	2.005	28.779	9.195	45.246	1.190	63.896	25.031	22.396	0.828	31.699	17.206
GPT-4o	38.540	2.418	26.782	8.979	45.946	1.348	57.281	21.631	14.330	0.532	31.304	17.278
Open-Source Models												
Qwen2.5-72B-Instruct	34.309	2.000	28.779	9.292	39.112	1.075	61.136	26.985	0.549	0.325	1.979	0.897
Llama-3.1-70B-Instruct	43.944	2.646	27.903	8.993	37.545	0.891	70.815	29.813	20.440	0.758	31.604	17.813
DeepSeek-67B-Chat	32.536	1.909	28.628	10.782	35.647	0.885	67.660	33.359	28.307	0.891	37.219	17.944
Yi-1.5-34B-Chat	30.743	1.823	28.335	9.195	35.364	0.808	73.561	35.490	13.620	0.434	36.778	22.790
Qwen2.5-32B-Instruct	26.332	1.980	22.348	5.261	21.336	0.729	49.157	20.704	11.566	0.869	15.608	7.984
DeepSeek-V2-Lite (15.7B)	16.686	0.974	28.771	16.806	31.458	0.744	68.524	35.404	4.804	0.153	36.846	20.562
Yi-1.5-9B-Chat	29.028	1.700	28.682	12.588	31.350	0.703	74.895	37.975	7.953	0.253	36.799	26.545
Llama-3.1-8B-Instruct	39.079	2.320	28.299	10.341	35.622	0.832	71.936	36.383	20.521	0.646	37.240	21.104
Qwen-2.5-Instruct-7B	4.291	0.285	24.933	14.156	41.203	0.925	74.862	37.975	19.477	0.612	37.289	20.796
FLAG-TRADER												
SmolLM2-135M-Instruct	34.342	2.429	23.913	10.872	50.394	1.362	64.004	37.975	45.511	1.734	30.903	24.440

[1] The Buy & Hold strategy is a passive investment approach commonly used as a baseline strategy, where an investor purchases stocks and holds onto them for an extended period regardless of market fluctuations.

[2] An upward arrow (↑) next to a metric indicates that higher values signify better performance, while a downward arrow (↓) indicates that lower values are preferable.

[3] The numbers highlighted in red indicate the best-performing outcomes for the corresponding metrics.

5.3 Experimental Results

- **FLAG-Trader achieves superior stock trading performance.** Unlike the baseline agent, which relies on an LLM-agentic framework, FLAG-TRADER undergoes an RL-based post-training phase. As shown in Table 1 and Table 2, FLAG-TRADER consistently outperforms the baseline across multiple metrics, demonstrating its stronger adaptability and optimization in diverse market environments.

- **Convergence to a (stable) optimal policy.** When using an LLM as the policy network in deep RL, the policy is jointly parameterized by the model's intrinsic parameters and the prompt. Although the initial prompts strongly influence policy generation during the first few training epochs, this effect diminishes over time. Consequently, the system converges to a relatively stable policy that becomes less sensitive to the initial prompts, indicating that RL training allows the LLM-based agent to refine its strategy and achieve a robust trading policy.

- **FLAG-TRADER enables small-scale models to surpass large-scale counterparts.** While increasing model size generally enhances financial decision-making and robustness— as seen with large proprietary models (e.g., GPT-o1-preview) in the baseline framework-FLAG-TRADER leverages an RL-based training pipeline to enable a 135M-parameter open-source model to outperform significantly larger models in financial trading tasks. This demonstrates that a well-designed training strategy can bridge or even surpass the performance gap typically associated with model scale.

6 Conclusion

In this paper, we introduced FLAG-TRADER, a novel framework that integrates LLMs with RL for financial trading. In particular, FLAG-TRADER leverages LLMs as policy networks, allowing for natural language-driven decision-making while benefiting from reward-driven optimization through RL fine-tuning. Our framework enables small-scale LLMs to surpass larger proprietary models by efficiently adapting to market conditions via a structured reinforcement learning approach. Through extensive experiments across multiple stock trading scenarios, we demonstrated that FLAG-TRADER consistently outperforms baseline methods, including LLM-agentic frameworks and conventional RL-based trading agents. These results highlight the potential of integrating LLMs with RL to achieve adaptability in financial decision-making.

Limitations and Potential Risk

Despite its promising results, FLAG-TRADER has several limitations. First, while our approach significantly enhances the decision-making ability of LLMs, it remains computationally expensive, particularly when fine-tuning on large-scale market datasets. Reducing computational overhead while maintaining performance is an important direction for future research. Second, financial markets exhibit high volatility and non-stationarity, posing challenges for long-term generalization. Future work should explore techniques such as continual learning or meta-learning to enhance model adaptability in evolving market conditions. Third, while FLAG-TRADER effectively integrates textual and numerical data, its reliance on structured prompts could introduce biases in decision-making. Improving prompt design or exploring retrieval-augmented methods may further enhance robustness. Lastly, real-world trading requires stringent risk management, and FLAG-TRADER currently optimizes for financial returns without explicitly incorporating risk-sensitive constraints. Extending the framework to integrate risk-aware objectives and dynamic portfolio optimization could provide more robust and practical financial trading solutions.

References

Dogu Araci. 2019. Finbert: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.

Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Misha Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. 2021. Decision transformer: Reinforcement learning via sequence modeling. *Advances in neural information processing systems*, 34:15084–15097.

Renato Arantes de Oliveira, Heitor S Ramos, Daniel Hasan Dalip, and Adriano César Machado Pereira. 2020. A tabular sarsa-based stock market agent. In *Proceedings of the First ACM International Conference on AI in Finance*, pages 1–8.

Xiang Deng, Vasilisa Bashlovkina, Feng Han, Simon Baumgartner, and Michael Bendersky. 2023. What do llms know about financial markets? a case study on reddit market sentiment analysis. In *Companion Proceedings of the ACM Web Conference 2023*, pages 107–110.

Xiu Gao and Laiwan Chan. 2000. An algorithm for trading and portfolio management using q-learning and sharpe ratio maximization. In *Proceedings of the international conference on neural information processing*, pages 832–837. Citeseer.

Ben Hambly, Renyuan Xu, and Huining Yang. 2023. Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3):437–503.

Allen H Huang, Hui Wang, and Yi Yang. 2023. Finbert: A large language model for extracting information from financial text. *Contemporary Accounting Research*, 40(2):806–841.

Yuling Huang, Xiaoxiao Wan, Lin Zhang, and Xiaoping Lu. 2024. A novel deep reinforcement learning framework with bilstm-attention networks for algorithmic trading. *Expert Systems with Applications*, 240:122581.

John Hull. 2007. *Risk Management and Financial Institutions*. John Wiley & Sons.

O Jangmin, Jongwoo Lee, Jae Won Lee, and Byoung-Tak Zhang. 2006. Adaptive stock trading with dynamic asset allocation using reinforcement learning. *Information Sciences*, 176(15):2121–2147.

Gyeeun Jeong and Ha Young Kim. 2019. Improving financial trading decisions using deep q-learning: Predicting the number of shares, action strategies, and transfer learning. *Expert Systems with Applications*, 117:125–138.

Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-Fang Li, Shirui Pan, and 1 others. 2023. Time-llm: Time series forecasting by reprogramming large language models. *arXiv preprint arXiv:2310.01728*.

Haohang Li, Yupeng Cao, Yangyang Yu, Shashidhar Reddy Javaji, Zhiyang Deng, Yueru He, Yuechen Jiang, Zining Zhu, Koduvayur Subbalakshmi, Guojun Xiong, and 1 others. 2024a. Investorbench: A benchmark for financial decision-making tasks with llm-based agent. *arXiv preprint arXiv:2412.18174*.

Yang Li, Yangyang Yu, Haohang Li, Zhi Chen, and Khaldoun Khashanah. 2023. Tradinggpt: Multi-agent system with layered memory and distinct characters for enhanced financial trading performance. *arXiv preprint arXiv:2309.03736*.

Yuan Li, Bingqiao Luo, Qian Wang, Nuo Chen, Xu Liu, and Bingsheng He. 2024b. Cryptotrade: A reflective llm-based agent to guide zero-shot cryptocurrency trading. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1094–1106.

Zhipeng Liang, Hao Chen, Junhao Zhu, Kangkang Jiang, and Yanran Li. 2018. Adversarial deep reinforcement learning in portfolio management. *arXiv preprint arXiv:1808.09940*.

Xiao-Yang Liu, Guoxuan Wang, and Daochen Zha. 2023. Fingpt: Democratizing internet-scale data for financial large language models. *arXiv preprint arXiv:2307.10485*.